

Exp No: 15

Code:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned char hexa = 0xFF;
```

```
    unsigned char hundreds, tens, units;
```

```
    hexa = hexa / 10;
```

```
    P0 = B;
```

```
    units = B;
```

```
    hexa = hexa / 10;
```

```
    hundreds = ACC;
```

```
    tens = B;
```

```
    P1 = B;
```

```
    P2 = ACC;
```

```
    while(1);
```

```
}
```

Output:

The screenshot displays the µVision IDE interface for a project named "exp-15". The main window shows the source code of "exp-15.c" with the following content:

```
#include <reg51.h>

void main(void)
{
    unsigned char hexa = 0xFF;
    unsigned char hundreds, tens, units;

    hexa = hexa / 10;
    P0 = B;
    units = B;

    hexa = hexa / 10;
    hundreds = ACC;
    tens = B;

    P1 = B;
    P2 = ACC;

    while(1);
}
```

The disassembly window shows the following instructions:

```
19: while(1);
C:0x081B 80FE SJMP C:081B
133: MOV R0,#IDATALEN - 1
C:0x081D 787F MOV R0,#0x7F
```

The registers window shows the current state of the registers:

Register	Value
R0	0x00
R1	0x00
R2	0x00
R3	0x00
R4	0x00
R5	0x00
R6	0x00
R7	0xFF
Sys	0x02
a	0x05
b	0x0A
sp	0x0A
sp_max	0x0A
dpr	0x0000
PC	0x081B
states	415
sec	0.00020750
pwr	0x01

The command window shows the program running with a code size limit of 2K. The call stack window shows the current function call stack:

Name	Location/Value	Type
MAIN	C:0x0800	